

UPWORK PORTFOLIO CASE STUDY

Circula

Operations, Inventory & Impact Reporting for the Circular Economy

Recycling and material recovery businesses in emerging markets run on paper notebooks, WhatsApp, and fragmented spreadsheets. Consequently, when funders, offtakers, and international regulators demand environmental and social impact proof, companies cannot defend their numbers. As co-founder and product designer, I designed Circula to bridge this gap directly from yard operations to audit-grade reporting.

ROLE & SCOPE

Co-founder & Product Designer

CLIENT / ORGANIZATION

Circula (Circular Economy Platform)

ENGAGEMENT & TIMELINE

Co-founder · 10 Months (2024–2025)

KEY DELIVERABLES

7-Module SaaS, Inventory State Machine, Invoicing, ESG Reports

6

Material Streams Tracked

4

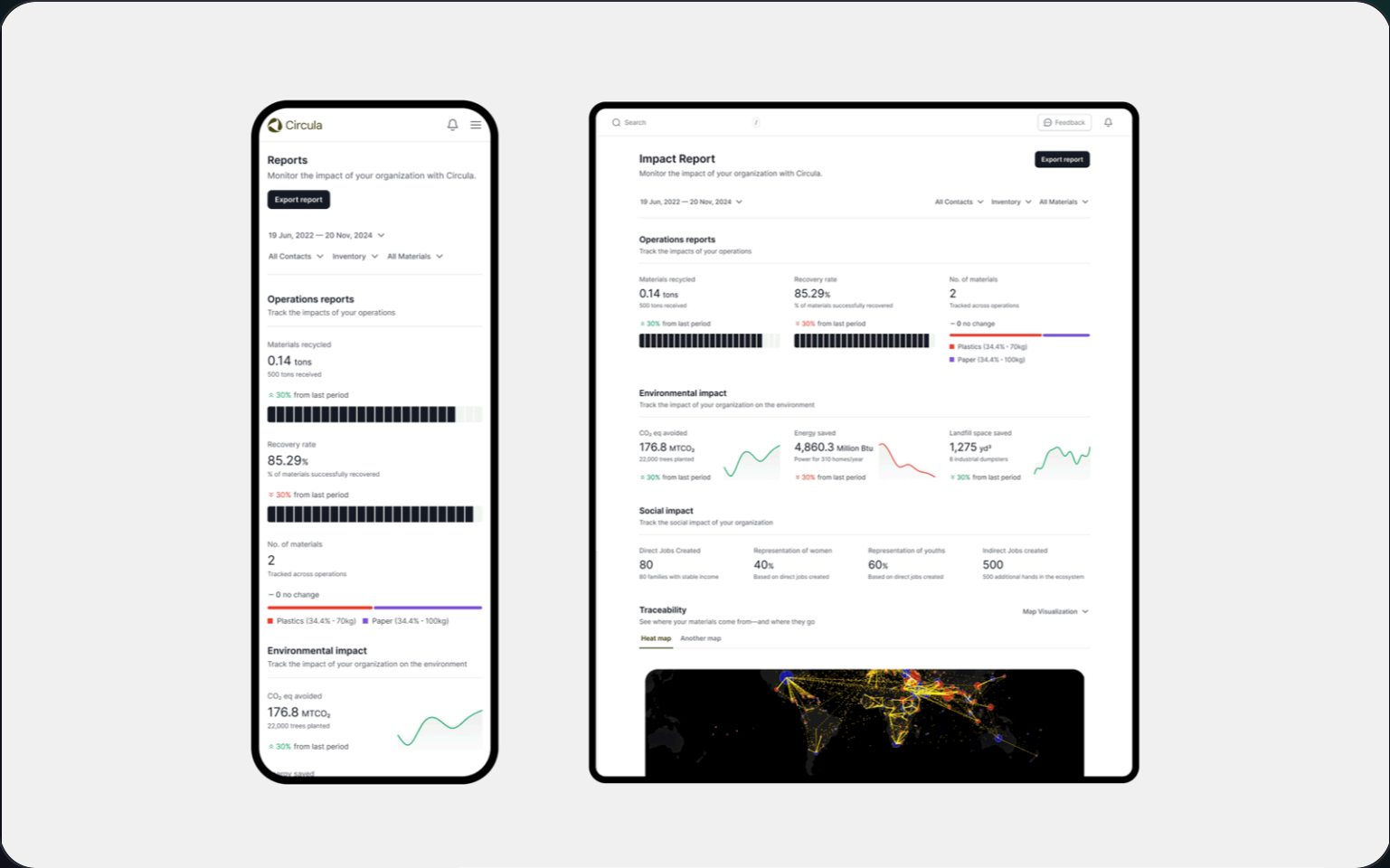
Invoicing Currencies

7

Operational Modules

100%

Automated Impact Calculation



■ PROBLEM FRAMING

# The Challenge & Friction Points

A single physical material undergoes continuous transformation: mixed plastic arrives from an informal collector, enters sorting, splits into recovered PET and residual waste, and only a portion becomes sellable stock against an invoice. If modelled too loosely, the recovery rate is fiction; if modelled too strictly, yard operators will bypass the system.

◆ Target User Segments & Stakeholders

Yard Supervisors

Managing intake scales, sorting lines, and baling crews; needed rapid kilogram entry with zero cognitive friction.

Operations Directors

Tracking multi-material inventory across warehouses and managing international offtaker contracts.

Funders & ESG Auditors

Demanding verifiable CO<sub>2</sub> avoided, landfill diversion, and direct/indirect green jobs data backed by immutable records.

■ STRATEGIC FOUNDATION

# Core Design Principles

01 Weigh Everything in Kilograms

Material weight is the unit the business runs on. Every screen measures in kilograms first and currency second, reflecting operational reality.

02 Waste is a First-Class Output

Recording an operation requires logging what was produced and what was wasted side-by-side. Recovery rates are only credible when loss is recorded at the moment of gain.

03 Impact is a Read, Not an Entry

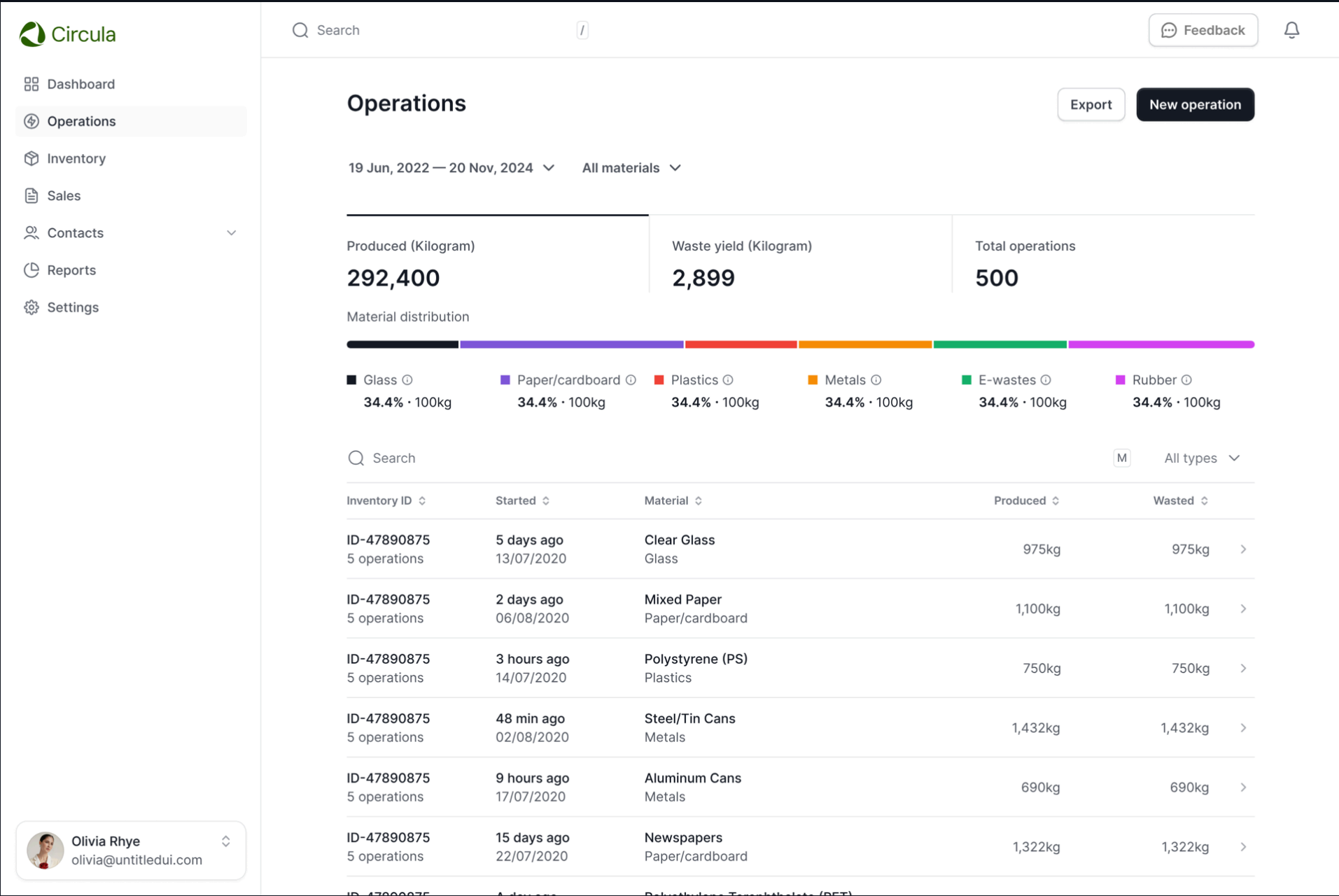
Nobody types their CO<sub>2</sub> figure in. Environmental and social impact are computed mathematically from operational records, making them legally defensible.

STRATEGIC INSIGHT

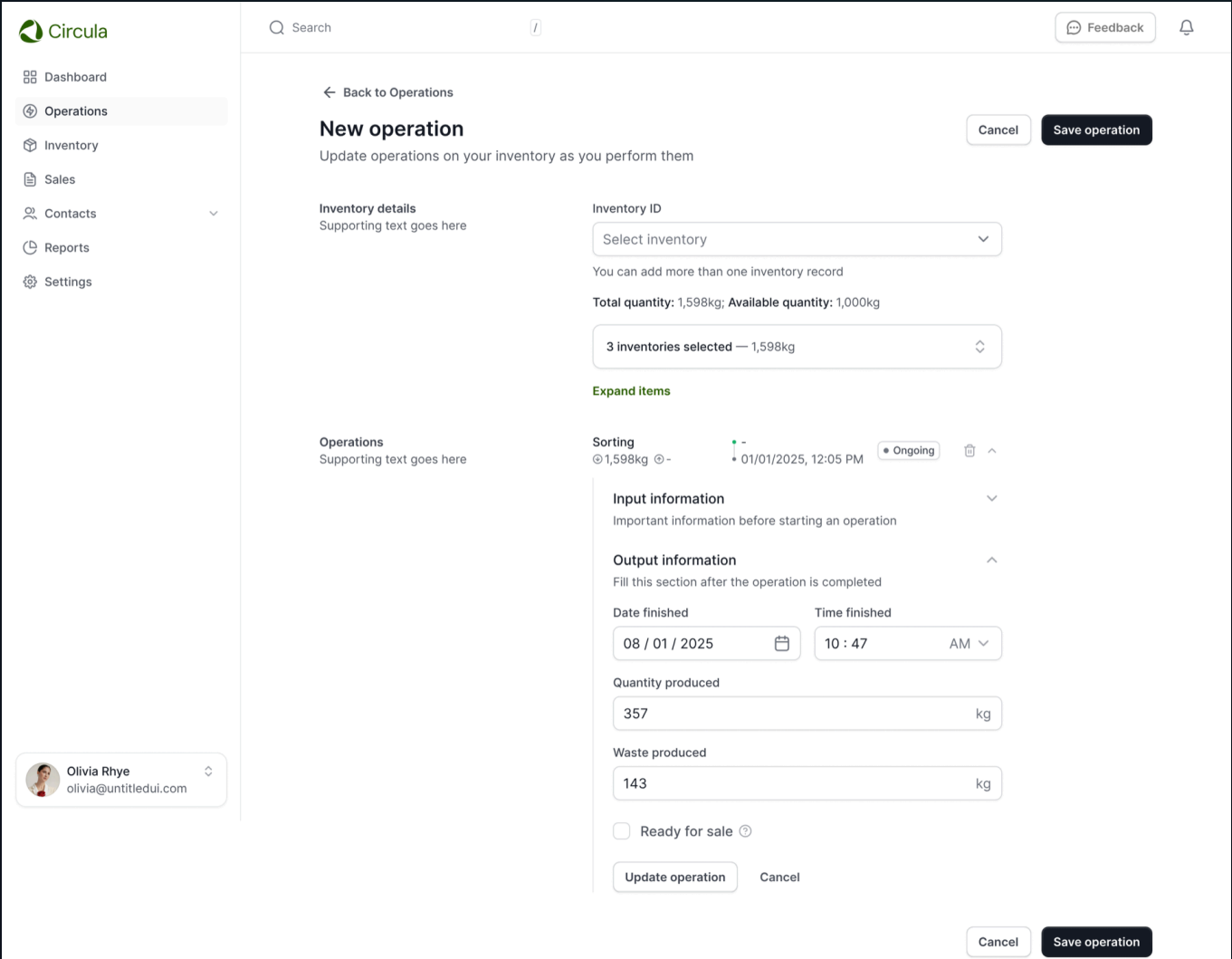
*"Impact reporting fails at the point of data entry, not at the point of reporting. Requiring waste logging alongside production output made downstream ESG reports completely defensible to international funders."*

■ SYSTEM BLUEPRINTS

# Material Transformation & Operational Lifecycle



Operations lifecycle tracking batch inputs, transformation stages, and output yields with real-time loss attribution.



New operation modal requiring input weight and residual waste before promoting output to sellable stock.

◆ Single Inventory Object

Anchored the platform on one unified inventory entity, with modules (sorting, baling, sales) representing distinct lifecycle states.

◆ Dual-Output Verification

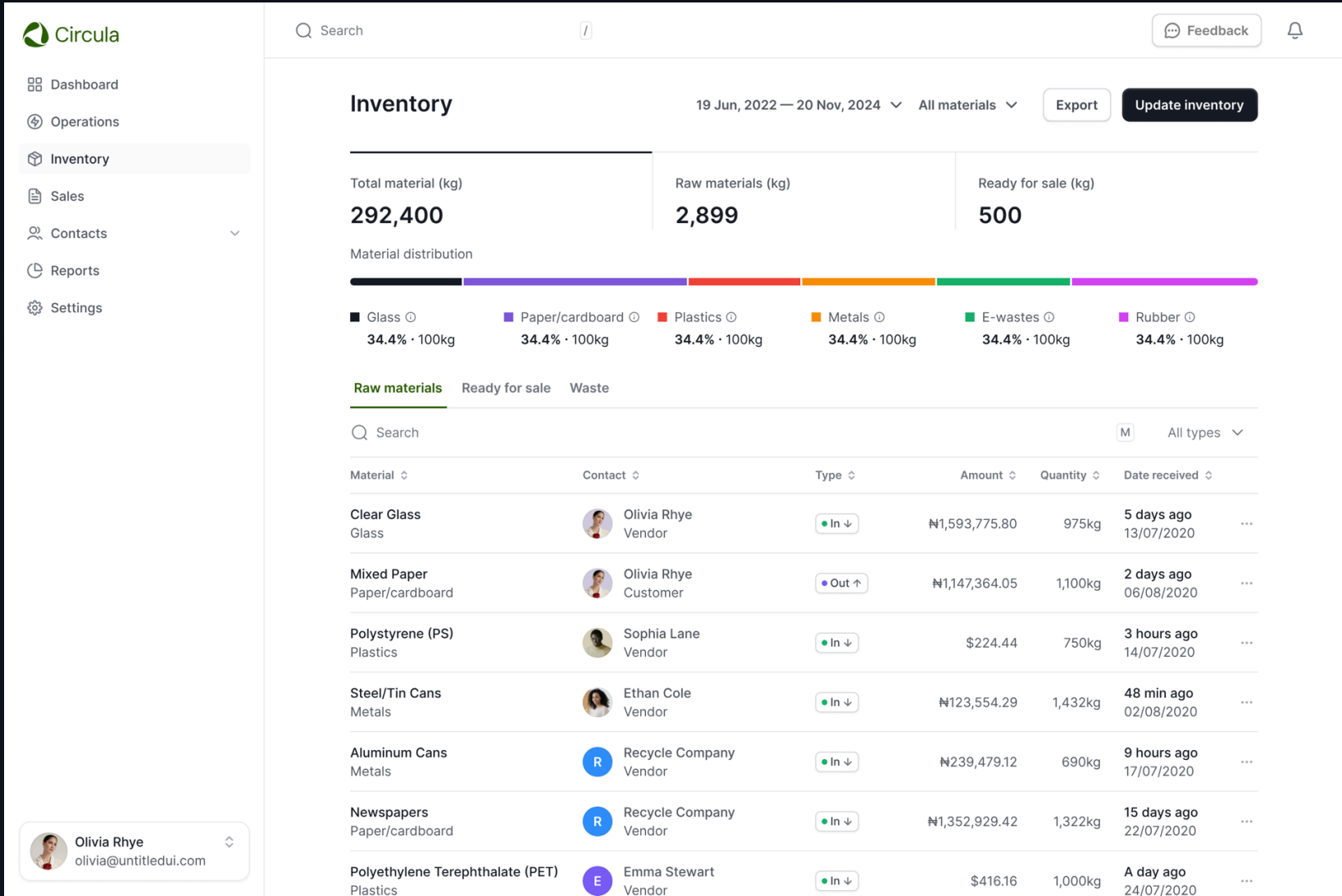
Operations require entering both recovered quantity and waste quantity, enforcing operational honesty on the ground.

◆ Multi-Currency Trade Engine

Supported local currency purchases (NGN) alongside foreign currency sales (USD, GBP, EUR) with live exchange rate logging.

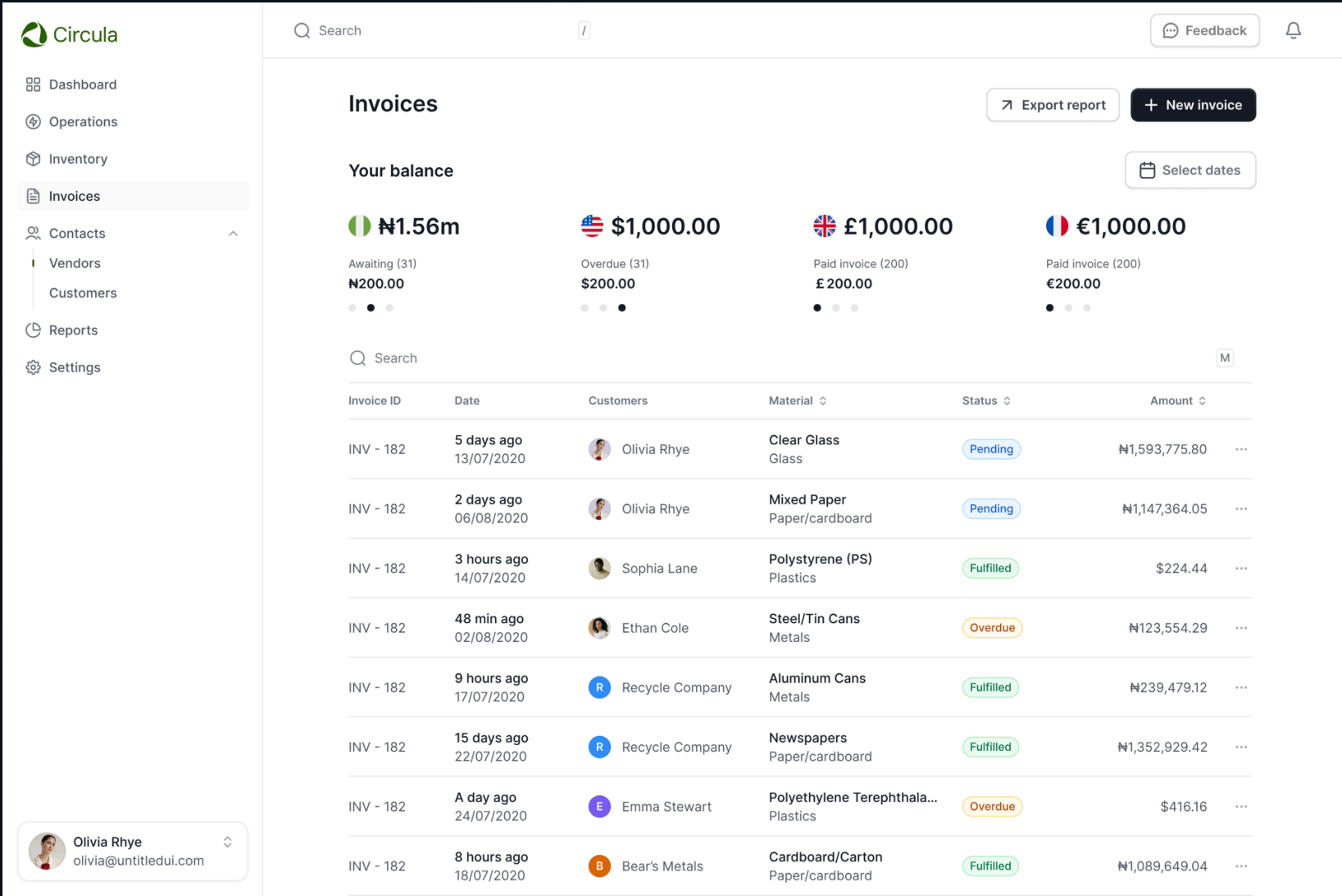
■ INTERFACE HIGHLIGHTS

# Core Product Experience & Touchpoints



● Inventory Across Three States

Raw materials, ready for sale, and waste live as tabs on one inventory page, allowing operators to understand stock progression without losing context.



● Multi-Currency Invoicing Engine

Balances held per currency with awaiting, overdue, and paid statuses, bridging local supply chain collections with international export orders.

■ SCALABLE FOUNDATIONS

# Tokens, States & Parity

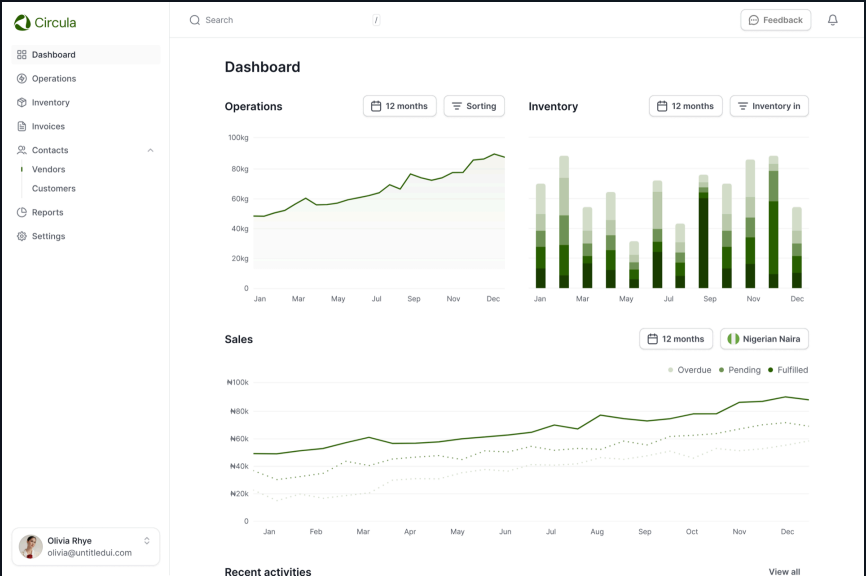
Restrained, dense design system built for high data throughput: crisp neutral ground, dark chrome navigation, and a single green accent reserved for brand and positive movement. Six fixed material stream hues (glass, paper, plastic, metal, e-waste, rubber) maintain 100% semantic consistency across distribution bars, tables, and reports.

### ◆ System Specifications

- ✓ Fixed 6-stream material color scale (Glass, Paper, Plastic, Metal, E-waste, Rubber)
- ✓ Unified dense table row anatomy across operations, inventory, and invoices
- ✓ Standardized status pills with clear semantic hierarchy
- ✓ Automated ESG calculation formulas translating kilograms to CO<sub>2</sub> metrics

### THE DESIGN ENGINEER ADVANTAGE

Bridging the gap between Figma and production front-end code (React, Next.js, TypeScript). Zero guesswork during developer handoff, pixel-perfect responsive execution, and bulletproof component architectures.



The screenshot shows the Circula Customers table with a sidebar navigation menu on the left. The table has columns for Name, Email, Phone number, Region, and Type. It lists 15 customers, including Olivia Rhye, Circula HQ, Sophia Lane, Ethan Cole, Recycle Company, Raid's Pickup, Emma Stewart, Bear's Metals, Isabella Reed, James Morgan, Hard Metal Pickup, Mia Green, Fire Recycler, Maya Thompson, Lia Harper, and Sophie Reynolds. Each row includes a status icon and a 'More' menu (three dots). A user profile for Olivia Rhye is visible in the bottom left corner.

| Name              | Email                         | Phone number   | Region             | Type       |
|-------------------|-------------------------------|----------------|--------------------|------------|
| Olivia Rhye       | olivia.rhye@circulahq.com     | +2348012345678 | Edo, NG            | Individual |
| Circula HQ        | admin@circulahq.com           | +2348012345679 | Lagos, NG          | Business   |
| Sophia Lane       | sophia.lane@circulahq.com     | +84753234734   | California, US     | Individual |
| Ethan Cole        | ethan.cole@circulahq.com      | +2348012345681 | Bavaria, DE        | Individual |
| Recycle Company   | recycle.company@circulahq.com | +2348012345682 | Tokyo, JP          | Business   |
| Raid's Pickup     | raid.pickup@circulahq.com     | +61480025873   | Ontario, CA        | Business   |
| Emma Stewart      | emma.stewart@circulahq.com    | +84234123908   | Queensland, AU     | Individual |
| Bear's Metals     | bears.metals@circulahq.com    | +2348012345685 | Rio de Janeiro, BR | Business   |
| Isabella Reed     | isabella.reed@circulahq.com   | +2348012345686 | Punjab, IN         | Individual |
| James Morgan      | james.morgan@circulahq.com    | +2348012345687 | Western Cape, ZA   | Individual |
| Hard Metal Pickup | hardmetals@circulahq.com      | +61488859548   | Auckland, NZ       | Business   |
| Mia Green         | mia.green@circulahq.com       | +2348012345689 | Scotland, GB       | Individual |
| Fire Recycler     | fire.recycler@circulahq.com   | +84667778889   | Kuala Lumpur, MY   | Business   |
| Maya Thompson     | maya.thompson@circulahq.com   | +2348012345691 | Moscow, RU         | Individual |
| Lia Harper        | lia.harper@circulahq.com      | +2348012345691 | Bangkok, TH        | Individual |
| Sophie Reynolds   | sophie.reynolds@circulahq.com | +2348012345691 | Seoul, KR          | Individual |

BUSINESS & USER IMPACT

# Shipped Outcomes & Measured Results

6

MATERIAL STREAMS TRACKED

4

INVOICING CURRENCIES

7

OPERATIONAL MODULES

100%

AUTOMATED IMPACT CALCULATION

◆ Validated Project Milestone

Shipped a comprehensive seven-module SaaS platform tracking six material streams from intake to sale. Enabled multi-currency invoicing across four currencies and delivered automated, defensible environmental and social impact reports.

◆ Senior Product Reflections

"Designing for the operational trade-off—making waste entry compulsory at the yard scale—was the pivotal decision. It created mild friction during recording but generated the only credible circular economy impact data in the sector."

JA

Jimmy Arikawe

Senior Product Designer & Design Engineer · Ex-Pentagram

AVAILABLE ON UPWORK

◆ End-to-End Product Design (0 to 1)

◆ Design Systems & Component Libraries

◆ Complex SaaS & Enterprise Dashboards

◆ Design Engineering (React / Next.js / TS)

🌐 jimmyarikawe.com

✉️ hi@jimmyarikawe.com

💼 linkedin.com/in/jimmyarikawe

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SLIDE 06 / 06